



The team that developed the TCE Cloud: (clockwise from top left)

K Balakrishnan, V Vivek Iyer, S Athma Sagar, V Ganapathi Subramanian, D Anand Kumar, M Varadharajan with Dr S Mercy Shalinie, and Karthick Seshadri

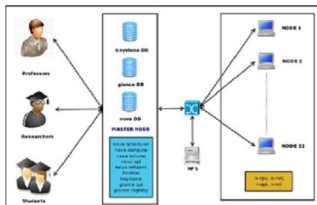


Figure 1: Architecture of the TCE cloud

metadata to instances), Nova Volume Controller (disk storage), Glance API, Glance Registry, Rabbit MQ and Nova Network (for communication with other nodes and with the virtual network). And these use external Python-based libraries such as Eventlet, Kombu and SQLAlchemy. Users are provided a dashboard using Horizon.

The TCE cloud

We have set up a cloud environment in the data centre of the Thiagarajar College of Engineering (TCE), to cater to the computational needs of research scholars and students. The following section details the architecture of the private cloud we set up.

We decided to build an OpenStack-based cloud, using 16 nodes of our 32-node cluster (Figure 2). Each node has two

cores, 3 GB of RAM and 80 GB of storage. Since OpenStack is constantly under development, we chose Diablo, a stable release, and chose Ubuntu 11.10 (Oneiric Ocelot) as the OS for this installation. The master node runs all services except for Swift. It also provides the Horizon dashboard, which will be the primary user interface. It has two physical network interfaces—one connected to the public college network, and the other connected to the private network of the cluster. This node also runs the MySQL database server required by various components.

Other nodes on the private network run Nova Compute, Nova Network, Nova API and Nova Volume. We started by reading OpenStack's documentation and tried a two-node installation. Once it was completed, we added the other nodes. Since not all packages in Ubuntu's repository were stable, we had to use DevStack for fetching stable packages, with a few modifications for the multi-node installation. Since only the master node was connected to the public network in our case, we chose SNAT for the master node, and used it as a gateway for Internet access on other nodes.

After we launched instances successfully using the dashboard, we tried uploading additional images. We tested Cirros, tylinux and Ubuntu 11.10 UEC. We could access our virtual machines from any location connected to the college network, including wireless access points.

Users can access and log in via the dashboard at *cloud.tcenet* from within the college. This component is named Horizon. The OpenStack Keystone component handles authentication. Once logged in, based on the type of user,